



Database Management System

Vehicle Service Booking & Tracking System

Smart Vehicle Servicing, Tracking & Billing Platform

React.js

Node.js

Express.js

MySQL

Firebase Cloud Messaging

TEAM MEMBERS

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Problem Statement

Traditional vehicle servicing is largely manual — this creates friction for customers and service centers alike.



HIGH

PROBLEM

Slot Discovery

Hard to find & book a suitable service slot at a convenient center.



HIGH

PROBLEM

Manual Scheduling

Paper-based scheduling causes delays and double-booking conflicts.



HIGH

PROBLEM

No Status Visibility

Customers don't know the current status of their vehicle service.



MEDIUM

PROBLEM

Technician Allocation

Assigning the right technician to the right job is hard to manage.



MEDIUM

PROBLEM

Spare Parts Visibility

Parts availability is not tracked, causing rework and delays.



MEDIUM

PROBLEM

Billing & History

Invoicing is slow and manual; past service records get lost.

Existing System

Most vehicle service centers still run on manual, disconnected processes with little digital support.



Manual Scheduling

Bookings are taken by phone or in person and logged into registers or spreadsheets.



Paper-Based Records

Job cards, service history and invoices are kept on paper and easily lost or misfiled.

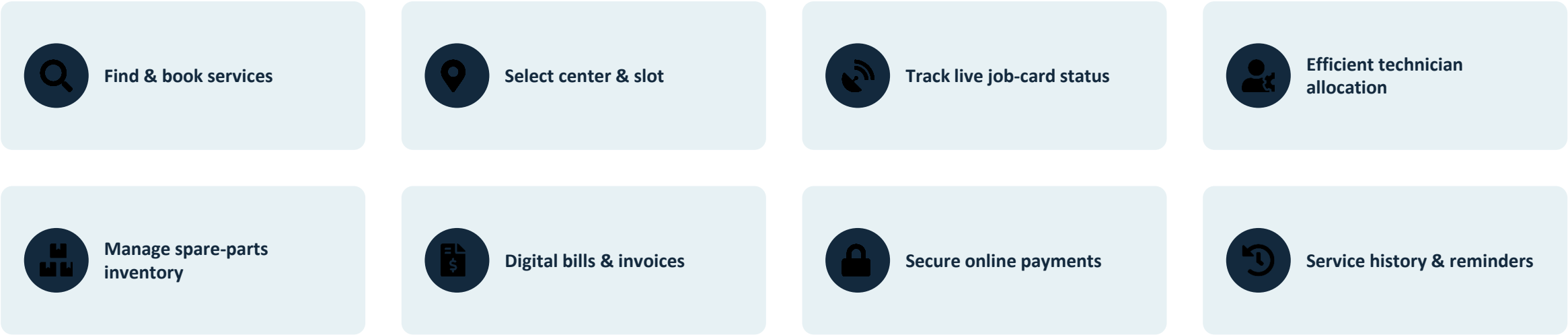


No Status Visibility

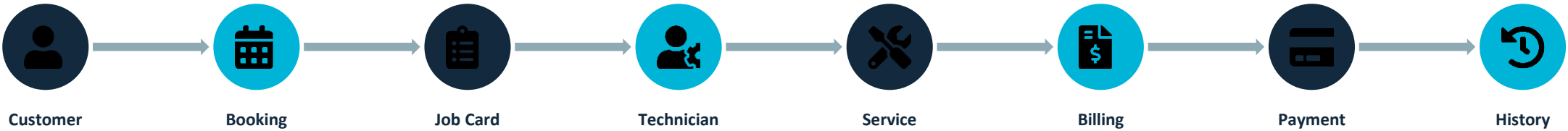
Customers have no way to check on their vehicle except by calling or visiting in person.

Proposed Solution

A centralized digital platform that connects customers, service centers, technicians and payments in one place.



END-TO-END WORKFLOW



Objectives

Core goals guiding the design and development of the platform.



Simplify Booking

Let customers browse services and book a slot online in a few clicks.



Enable Live Tracking

Give customers real-time visibility into their vehicle's service status.



Streamline Operations

Help service centers manage technicians, parts and schedules efficiently.



Digitize Billing

Auto-generate accurate invoices and support secure online payments.



Maintain Service History

Keep a complete digital record of every vehicle's past services.



Improve Communication

Send real-time notifications and maintenance reminders via FCM.

Project Scope

What phase one of the platform covers, and what is planned for later.

In Scope

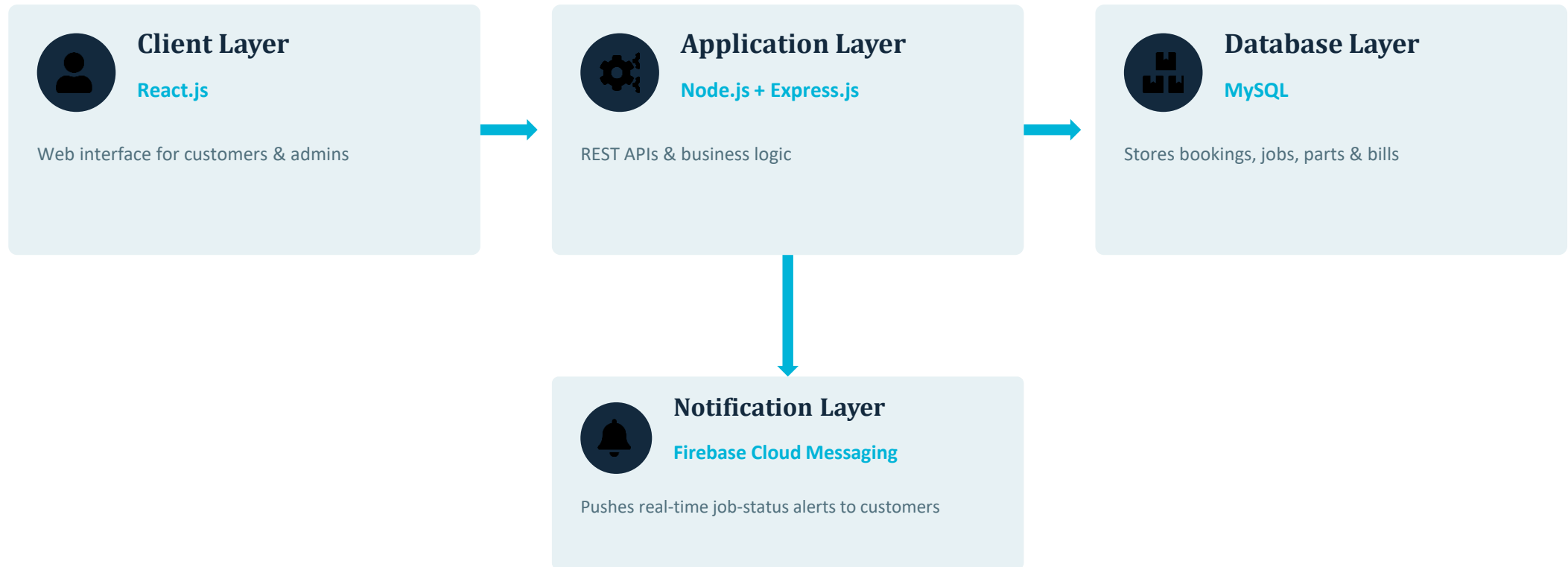
-  Customer registration & vehicle management
-  Service catalog & online slot booking
-  Digital job-card & technician allocation
-  Spare-parts inventory tracking
-  Digital billing & online payments
-  Service history & maintenance reminders

Out of Scope (Phase 1)

-  Dedicated native mobile application
-  AI-based service recommendations
-  IoT-based vehicle health monitoring
-  Live GPS vehicle tracking
-  Multi-language support
-  Third-party fleet / insurance integrations

Basic Architecture

A layered client-server architecture connecting the front end, backend services, database and real-time notifications.



Key Features



Service Catalog & Booking

Browse & book services in a few clicks.



Live Job-Card Tracking

Real-time visibility of service progress.



Technician Allocation

Assigns available technicians efficiently.



Spare Parts Inventory

Tracks required & available parts.



Digital Billing & Invoice

Auto-generates clear, accurate bills.



Online Payment

Secure payments powered by Razorpay.



Service History

Complete record of past services.



Maintenance Reminders

Alerts for upcoming service needs.



Real-Time Notifications

Instant updates via Firebase (FCM).

System Workflow



User Modules



Customer Module

Book, track & manage your service

- ✓ Register / Login
- ✓ Manage Vehicles
- ✓ Browse Services
- ✓ Book Service Slots
- ✓ Track Job-Card Status
- ✓ Receive Notifications
- ✓ Make Payments
- ✓ View Invoices
- ✓ Service History
- ✓ Maintenance Reminders

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Self-service capabilities

Everything a customer needs, from booking to payment.



Service Center / Admin

Operate, allocate & fulfil services

- ✓ Manage Service Catalog
- ✓ Manage Bookings
- ✓ Create Job Cards
- ✓ Allocate Technicians
- ✓ Update Service Status
- ✓ Manage Spare Parts
- ✓ Generate Bills / Invoices
- ✓ View Service History

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Operational capabilities

One dashboard to run the entire service center.

Unique Features & Innovation

What sets this project apart from a basic vehicle-booking application.



Live Job-Card Status

See the current stage of servicing.



Technician Allocation

Assigns available technicians to jobs.



Spare Parts Tracking

Monitors required & available parts.



Digital Billing

Generates clear bills & invoices.



Online Payment

Convenient payment via upi.



Service History

Stores previous service records.



Smart Reminders

Helps remember future maintenance.



Real-Time Notifications

Uses FCM for important updates.

Expected Benefits & Future Scope

Expected Benefits

-  Faster, easier service booking
-  Better service center organization
-  Improved customer transparency
-  Reduced manual work
-  Efficient technician utilization
-  Better spare-parts management
-  Faster billing & payment
-  Easy access to service history

Future Scope

-  AI-based service recommendations
-  Predictive vehicle maintenance
-  IoT-based vehicle health monitoring
-  GPS-based vehicle / service tracking
-  Dedicated mobile application
-  Automated support chatbot

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Growth opportunities

Roadmap items to extend the platform beyond phase one.



Conclusion

The Vehicle Service Booking & Tracking System provides a centralized platform for booking, managing, tracking and completing vehicle services — connecting customers, service centers, technicians, inventory, billing and payments in one system.

“Making Vehicle Servicing Smarter, Faster & More Transparent.”

Thank You

Questions?